

A Circle Counterexample to Exponential Localization of Restricted Surface-Spline Lagrange Functions

Run `fa_banach_001`, lane 1

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Status

This packet gives a candidate counterexample to Conjecture 1 of Hangelbroek–Narcowich–Sun–Ward [1] as literally stated. The counterexample uses the allowed endpoint $d = m = 1$, hence the restricted surface spline on S^1 . It disproves exponential localization; it does not disprove the separate Holder estimate or the approximation consequences of the conjecture.

Source Problem

The source is arXiv:1005.2424, Section 6, PDF page 20. It defines restricted surface-spline spaces on S^d and conjectures that their Lagrange functions form a local basis in the sense of Assumption 3.4 and satisfy Assumption 3.5. The source page is reproduced below.

For centers $\Xi \subset \mathbb{S}^d$, define kernel spaces

$$S(\varphi_m, \Xi) := \left\{ \sum_{\xi \in \Xi} A_\xi \varphi_m(\cdot, \xi) + P \mid P \in \Pi_{[m-d/2]} \text{ and } \sum_{\xi \in \Xi} A_\xi Q(\xi) = 0, \forall Q \in \Pi_{[m-d/2]} \right\}$$

generated by shifts of the restricted surface splines

$$\varphi_m(x, y) := \begin{cases} (1 - \langle x, y \rangle)^{m-d/2} \log(1 - \langle x, y \rangle) & d \text{ even} \\ (1 - \langle x, y \rangle)^{m-d/2} & d \text{ odd} \end{cases},$$

and spherical harmonics of degree $[m - d/2]$ or less. Such kernels are known to be of the form $G_\beta + G_\beta * \psi$, with $\beta = 2m$ [12] (3.6)] and thus have a precise, theoretical error analysis (similar to the case of $\kappa_{2, \mathbb{S}^2}$ of the previous section). They also invert an elliptic differential operator of the form $\prod_{j=1}^m (\Delta - r_j)$, with $r_1, \dots, r_m$ a sequence of real numbers [7 Lemma 3.4], as such, they are very similar to the kernels $\kappa_{m, \mathbb{S}^d}$, which also invert elliptic differential operators of order $2m$.

Conjecture 1. *If Ξ is quasiuniform with mesh ratio ρ , the Lagrange functions $\chi_\xi \in S(\varphi_m, \Xi)$ are a local basis (in the sense of Assumption 3.4) and satisfy the Hölder continuity assumption, Assumption 3.5*

The validity of this conjecture would have several immediate consequences:

- [A] The associated Lebesgue constants for interpolation are bounded by a constant depending only on ρ (and not on $\#\Xi$),
- [B] Interpolation provides precise L_∞ approximation rates. For functions $f \in C^s$, $s \leq 2m$, one has
$$\|f - I_\Xi f\|_\infty = \mathcal{O}(h^s).$$
- [C] The Lagrange basis satisfies the stability condition (3.1),
- [D] The L_p norms of the L_2 projector T_Ξ associated with these spaces are bounded with constants depending only on ρ .
- [E] The L_2 projector provides precise L_p approximation rates. I.e., approximation order s for functions having L_p smoothness s , for $s \leq 2m$ (as determined by inclusion in Bessel potential H_{2m}^p or Besov spaces $B_{p, \infty}^s$):
$$\|f - T_\Xi f\|_p = \mathcal{O}(h^s)$$

A more ambitious problem would be to prove the previous for all kernels of the form

$$\phi_\beta = G_\beta + G_\beta * \psi, \quad \beta > d,$$

which includes the compactly supported SBFs, as well as most other SBFs of finite smoothness.

Conjecture 2. *If Ξ is quasiuniform with mesh ratio ρ , the Lagrange functions $\chi_\xi \in S(\phi_\beta, \Xi)$ are a local basis (in the sense of Assumption 3.4) and satisfy Assumption 3.5*

For a quasiuniform family, Assumption 3.4 requires constants $C_1, \nu > 0$, independent of the number of centers, such that

$$|\chi_\xi(x)| \leq C_1 \exp\left(-\nu \frac{\min(d(x, \xi), r_M)}{q}\right), \quad (1)$$

where q is the separation radius and r_M is the injectivity radius.

Counterexample Theorem

Theorem 1. *Conjecture 1 of [1] is false as stated. More precisely, take $d = m = 1$, identify S^1 with $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, and use the restricted surface spline*

$$\kappa(\theta) = \sqrt{1 - \cos \theta} = \sqrt{2} |\sin(\theta/2)|.$$

For every odd $N \geq 5$, let

$$\Xi_N = \{\theta_j = 2\pi j/N : 0 \leq j < N\}$$

and let χ_N be the Lagrange function centered at 0 in the source space $S(\kappa, \Xi_N)$, whose auxiliary space is $\Pi_1 = \text{span}\{1, \cos \theta, \sin \theta\}$. Then

$$\chi_N(\pi) = \frac{1}{N} \left(\frac{2 \cos(\pi/N) - 1}{\cos(\pi/(2N))} - 1 \right), \quad (2)$$

and consequently

$$\lim_{\substack{N \rightarrow \infty \\ N \text{ odd}}} N^3 \chi_N(\pi) = -\frac{7\pi^2}{8}. \quad (3)$$

In particular, the exponential localization estimate (1) fails for the quasiuniform family $(\Xi_N)_{N \text{ odd}}$.

Proof Intuition

Uniform nodes force the low-frequency polynomial part of the cardinal function: the constant, cosine, and sine moments of the interpolation data give $p_N = (1 + 2 \cos \theta)/N$. The remaining part is a sum of translates of $\sqrt{2} |\sin(\theta/2)|$. Between nodes every translate solves the same homogeneous ODE $u'' + u/4 = 0$. On the grid cell centered at the antipode, the cardinal function vanishes at both endpoints, so this ODE can be solved exactly. The low-frequency term is almost, but not exponentially, cancelled: the residual is of order N^{-3} . Since the grid scale is $q = \pi/N$, the conjecture would require an e^{-cN} residual instead.

Proof

Proof. For $d = m = 1$, the kernel and auxiliary space in the displayed definition of Conjecture 1 are exactly

$$\kappa(\theta) = \sqrt{1 - \cos \theta}, \quad \Pi_1 = \text{span}\{1, \cos \theta, \sin \theta\}.$$

The usual augmented interpolation system is nonsingular on the uniform grids used below. One quick verification is the Fourier series

$$\kappa(\theta) = \frac{2\sqrt{2}}{\pi} - \frac{4\sqrt{2}}{\pi} \sum_{n \geq 1} \frac{\cos(n\theta)}{4n^2 - 1}.$$

Thus $-\kappa$ has strictly positive Fourier coefficients in every mode $|n| \geq 2$. On coefficient vectors satisfying the three side conditions, the modes $0, \pm 1$ vanish, so the quadratic form is strict. Multiplying a kernel by -1 does not change its interpolation space or its Lagrange functions.

Write the Lagrange function centered at 0 as

$$\chi_N(\theta) = \sum_{j=0}^{N-1} A_j \kappa(\theta - \theta_j) + p_N(\theta), \quad p_N(\theta) = a + b \cos \theta + c \sin \theta, \quad (4)$$

with

$$\sum_j A_j = \sum_j A_j \cos \theta_j = \sum_j A_j \sin \theta_j = 0 \quad (5)$$

and $\chi_N(\theta_\ell) = \delta_{0\ell}$.

We first determine p_N without determining any A_j . Sum the cardinal equalities over ℓ . The row sum $\sum_\ell \kappa(\theta_\ell - \theta_j)$ is independent of j , so the kernel contribution vanishes by the first side condition. Discrete orthogonality on the uniform grid gives

$$1 = \sum_\ell \chi_N(\theta_\ell) = Na,$$

hence $a = 1/N$. Next multiply the cardinal equalities by $\cos \theta_\ell$ and sum. After the change of index $r = \ell - j$, the kernel contribution is a fixed linear combination of $\sum_j A_j \cos \theta_j$ and $\sum_j A_j \sin \theta_j$, hence vanishes by (5). Therefore

$$1 = \sum_\ell \delta_{0\ell} \cos \theta_\ell = b \sum_\ell \cos^2 \theta_\ell = \frac{N}{2} b.$$

The analogous sine-weighted identity gives $c = 0$. Thus

$$p_N(\theta) = \frac{1 + 2 \cos \theta}{N}. \quad (6)$$

Now let N be odd. The point π is the midpoint of the grid cell with endpoints

$$\pi - \frac{\pi}{N}, \quad \pi + \frac{\pi}{N}.$$

Neither endpoint is the center 0, so χ_N vanishes at both. On the open cell there is no kernel center, and every translate of κ obeys

$$v'' + \frac{1}{4}v = 0.$$

Consequently $u_N = \chi_N - p_N$ obeys the same ODE there. Put $t = \theta - \pi$. By (6), the two endpoint values of p_N are equal, so the two endpoint values of u_N are equal. The sine term in the general solution of the ODE therefore vanishes, and

$$u_N(\pi + t) = B_N \cos(t/2).$$

At $t = \pi/N$, the cardinal condition and (6) give

$$B_N \cos\left(\frac{\pi}{2N}\right) = -p_N\left(\pi + \frac{\pi}{N}\right) = \frac{2 \cos(\pi/N) - 1}{N}.$$

Since $p_N(\pi) = -1/N$, evaluating at $t = 0$ proves (2).

Let $x = \pi/N$. Taylor expansion at 0 yields

$$\frac{2 \cos x - 1}{\cos(x/2)} = 1 - \frac{7}{8}x^2 + O(x^4).$$

Substitution into (2) proves (3).

Finally, the uniform grid has fill distance and separation radius

$$h_N = q_N = \frac{\pi}{N},$$

so its mesh ratio is 1. On S^1 , $d(0, \pi) = r_M = \pi$. If Assumption 3.4 held uniformly for this quasiuniform family, it would give

$$|\chi_N(\pi)| \leq C_1 \exp\left(-\nu \frac{\pi}{q_N}\right) = C_1 e^{-\nu N}.$$

But (3) gives $|\chi_N(\pi)| \sim 7\pi^2/(8N^3)$, a contradiction. $\square$

Verification Notes

The script `code/verify_antipodal_formula.py` independently solves the full augmented interpolation system for several odd values of N , compares the computed antipodal value with (2), and displays $-N^3 \chi_N(\pi)$ approaching $7\pi^2/8$. This is only a check of the algebra; the ODE calculation above is the proof.

The source statement contains no restriction $d \geq 2$. If such a restriction was intended but omitted, Theorem 1 should be read as an endpoint obstruction to the literal conjecture, not as a result about $S^3, S^5, \dots$

Relation to Later Literature

Hangelsbroek–Narcowich–Ward [2], posted after the source paper, cites [1] and treats the same kernels. Its Theorem 5.3 proves the exact exponential estimate for the even-dimensional spherical case. Its Theorem 5.5 proves, in the general case, a bound with a polynomial floor,

$$|\chi_\xi(x)| \leq C \rho^{m-d/2} \max \left\{ e^{-\nu \min(d(x, \xi), r_M)/h}, h^{2m} \right\},$$

as well as the Holder estimate. The discussion preceding Theorem 5.3 says that exponential decay on odd-dimensional spheres was believed to hold and that the slower bound might be an artifact of the proof. The exact S^1 formula above shows that pure exponential decay cannot hold in the full odd-dimensional formulation.

The later paper also proves bounded Lebesgue constants, L_p -stability of the Lagrange basis, boundedness of the L_2 -projector on every L_p , and the advertised approximation rates. Hence the counterexample targets only the strong exponential clause, not consequences [A]–[E].

Novelty Search and Limitations

The run registry and the solution, attempt, and proof-gap indexes were searched for arXiv:1005.2424, the paper title, restricted surface splines, Lagrange localization, and the circle

case. No duplicate was present. Bounded web searches on July 22, 2026 used the exact conjecture phrase and combinations of restricted surface splines, S1, odd-dimensional spheres, Lagrange functions, and exponential decay. They found the source, [2], and adjacent localization papers, but no prior counterexample or the antipodal formula (2).

This packet does not address Conjecture 2, the $SO(3)$ Conjecture 3, or a version of Conjecture 1 restricted to odd dimensions at least 3. It also does not contradict the Holder half of Conjecture 1 or any of the known stability and approximation theorems.

References

- [1] T. Hangelbroek, F. J. Narcowich, X. Sun, and J. D. Ward, *Kernel Approximation on Manifolds II: The L_∞ -norm of the L_2 -projector*, arXiv:1005.2424; SIAM J. Math. Anal. 43 (2011), 662–684.
- [2] T. Hangelbroek, F. J. Narcowich, and J. D. Ward, *Polyharmonic and Related Kernels on Manifolds: Interpolation and Approximation*, arXiv:1012.4852; Found. Comput. Math. 12 (2012), 625–670.